20A, 600V Low Loss Surface Mount Rectifier

FEATURES

- Lead for automated placement
- Low forward voltage drop
- High surge current capability
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

APPLICATIONS

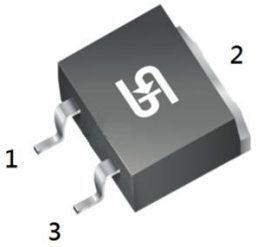
- General purpose
- Switching mode power supply (SMPS)
- On board charger
- Totem-pole PFC

MECHANICAL DATA

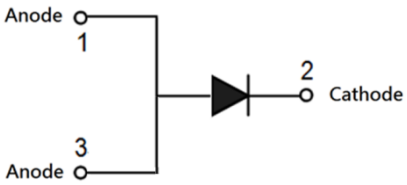
- Case: D^2PAK-D
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: As marked
- Weight: 1.41g (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	20	A
V_{RRM}	600	V
I_{FSM}	390	A
$T_{J\ MAX}$	175	°C
Package	D^2PAK-D	

Pb-Free RoHS COMPLIANT HALOGEN FREE



D^2PAK-D



ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)				
PARAMETER		**SYMBOL**	**VALUE**	**UNIT**
Repetitive peak reverse voltage		V_{RRM}	600	V
Reverse voltage, total rms value		$V_{R(RMS)}$	420	V
Forward current		I_F	20	A
Surge peak forward current single half sine-wave superimposed on rated load	t = 8.3ms	I_{FSM}	390	A
	t = 1.0ms		900	
Junction temperature		T_J	-55 to +175	°C
Storage temperature		T_{STG}	-55 to +175	°C

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-lead thermal resistance	$R_{\Theta JL}$	1.7	°C/W
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	6.7	°C/W
Junction-to-case thermal resistance	$R_{\Theta JC}$	1.3	°C/W

Thermal Performance Note: Mounted on Heat sink with 4" x 6" x 0.25" Al-Plate

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER	CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage (1)	I_F = 10A, T_J = 25°C	V_F	0.88	-	V
	I_F = 20A, T_J = 25°C		0.93	1.0	V
	I_F = 10A, T_J = 125°C		0.75	-	V
	I_F = 20A, T_J = 125°C		0.82	-	V
Reverse current @ rated V_R (2)	T_J = 25°C	I_R	-	10	μA
	T_J = 125°C		0.5	-	μA
Junction capacitance	1MHz, V_R = 4.0V	C_J	152	-	pF
Reverse recovery time	I_F = 0.5A, I_R = 1.0A, I_{rr} = 0.25A	t_{rr}	1060	-	ns

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE	PACKAGE	PACKING
PLDS20J	D^2PAK-D	800 / Tape & Reel

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

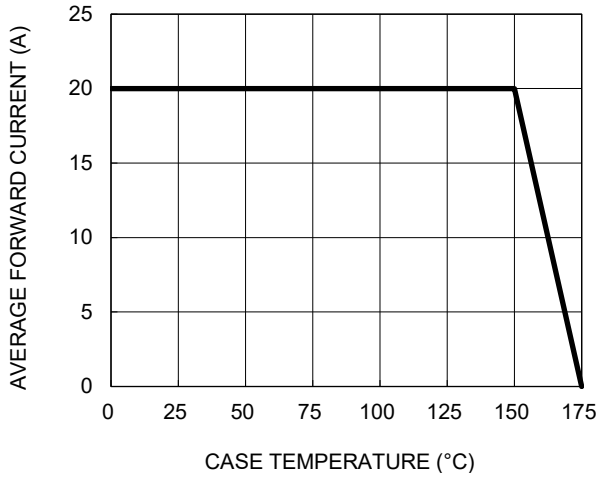


Fig.2 Typical Junction Capacitance

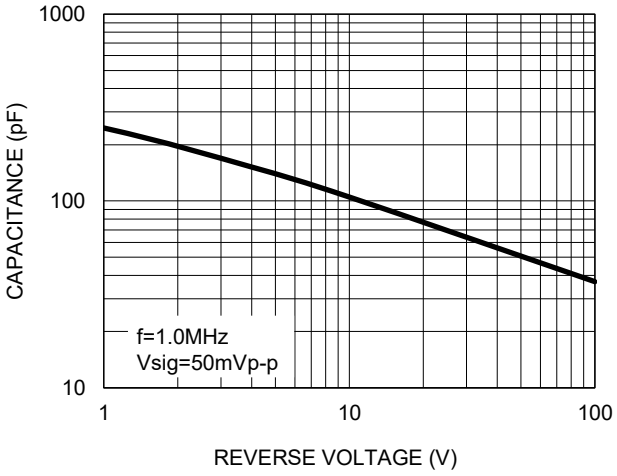


Fig.3 Typical Reverse Characteristics

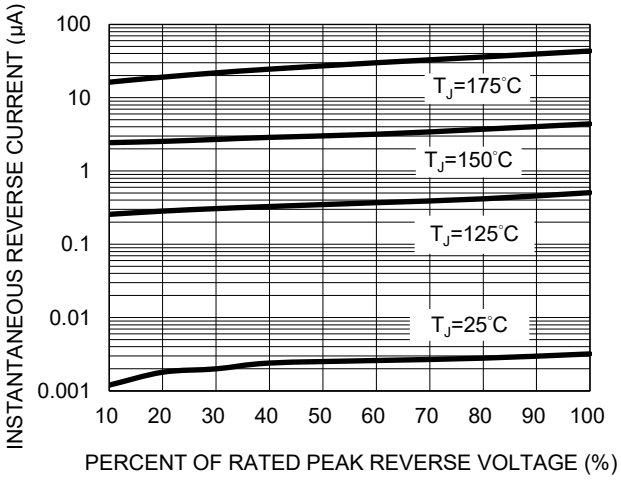


Fig.4 Typical Forward Characteristics

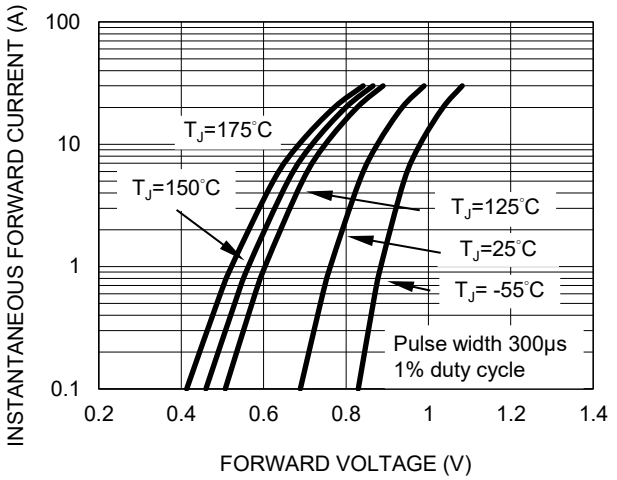
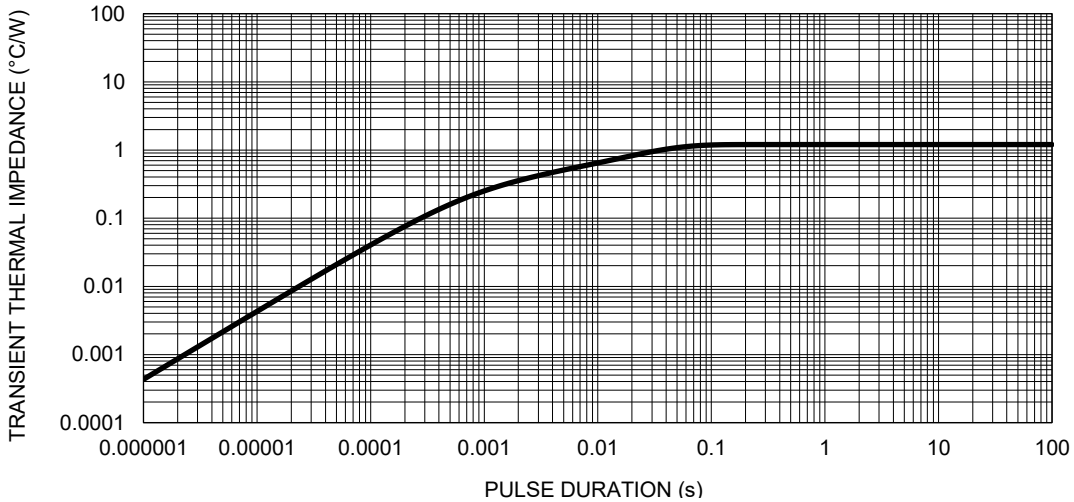
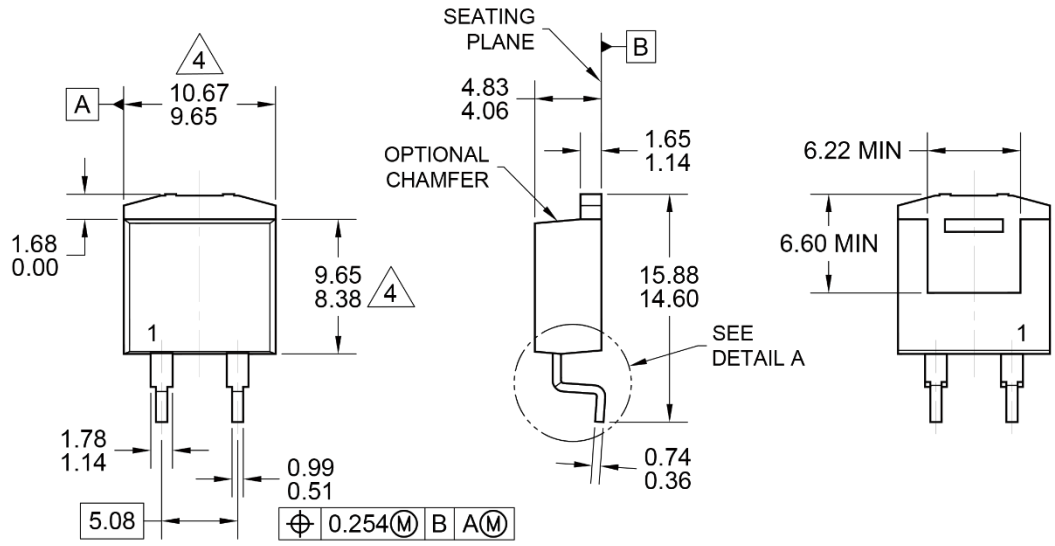


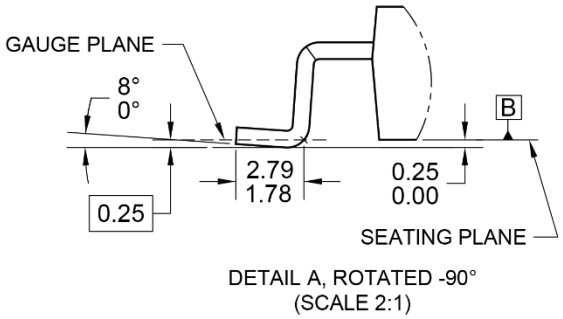
Fig.5 Typical Transient Thermal Impedance



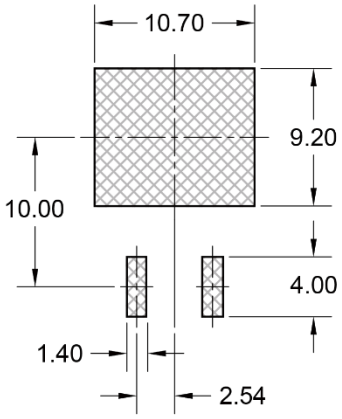
PACKAGE OUTLINE DIMENSIONS

D²PAK-D

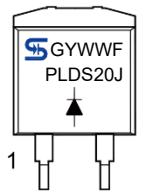




DETAIL A, ROTATED -90°
(SCALE 2:1)



SUGGESTED PAD LAYOUT



MARKING DIAGRAM

G = GREEN COMPOUND
YWW = DATE CODE
F = FACTORY CODE

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. THE PACKAGE OUTLINE REFERENCE: JEDEC TO-263, VARIATION AB, ISSUE F.
4. MOLDED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH. THESE DIMENSIONS ARE MEASURED AT THE OUTERMOST EXTREME OF THE PLASTIC BODY.
5. DWG NO. REF: HQ2SD07-D2PAKD-066 REV B.

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